Scenario 3: Social Media Platform for a Startup

Context:
You are working with a startup to design **ChatVibe**, a new social media app aimed at teenagers. The app allows users to:

1. Create profiles.
2. Share posts (text, images, short videos).
3. Like and comment on others' posts.
4. Send direct messages.

The startup's goals:

- **Investors**: Demand that the app launches within **four months** to gain traction before competitors roll out similar features.
- **Beta Testers**: Have requested advanced features like **live video streaming** and **group chats**, claiming these are essential to make the app stand out.
- **CTO**: Warns that adding these features now will increase complexity and delay the launch. Recommends starting with a Minimal Viable Product (MVP) focusing on basic features.
- **Marketing Team**: Says the app must support a **viral growth strategy**, handling up to **5 million users in the first six months**. They also emphasize that the app's performance (e.g., loading posts, sending messages) must remain fast to retain users.

Budget constraints:

1. The startup has limited funding and cannot afford to over-provision servers.
2. The engineering team consists of only **six developers** with varying levels of experience.

Challenge for Junior Architects:

1. What tradeoffs must be made to balance time-to-market, feature richness, and scalability?
2. Should you launch the MVP or delay to add advanced features? What risks does each approach pose?
3. How would you design the system to ensure scalability while staying within budget constraints? What technologies or design patterns would you use?
